Day-20

November-13.

Spanning Tree Protocol (Part-1)

Network Redundancy

(अहहा bigryo bhnye backup rakhelo hunu parxa, so it can perform = redundancy)

Modern networks are expected to run 24 hours a day, 7 days a week, 365 days a year. Even a short downtime can be disastrous for a business.

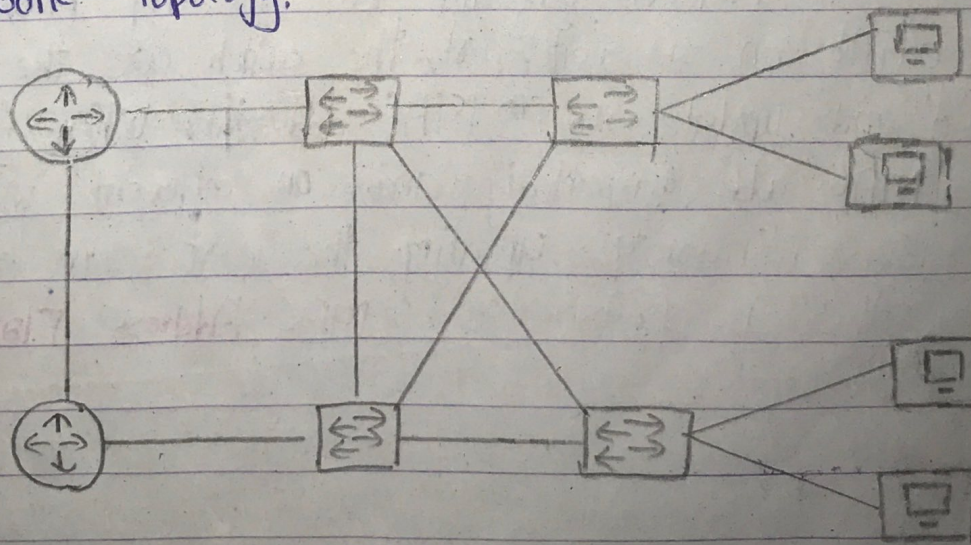
- If one network component fails, we must ensure that other components will take over with little or no downtime.

- As much as possible, we must implement redundancy at every possible ~~poi~~ point in the network. Redundancy is essential part of network design. A network that isn't redundant is simply not acceptable.

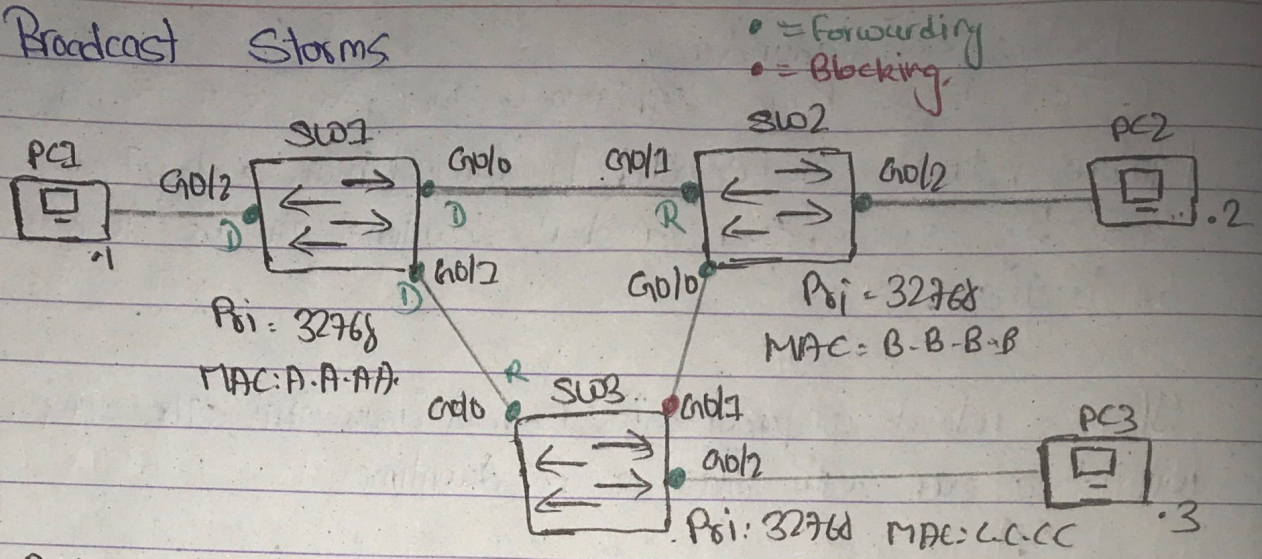
NOTE: Most PCs only have a single network interface card (NIC), so they can only be plugged into a single switch. However, important servers typically have multiple NICs, so they can be plugged into multiple switches for redundancy.

There are many protocols that are used to enable network redundancy ~~throughout this course~~, and spanning tree is one of them.

Network Topology.



Broadcast Storms



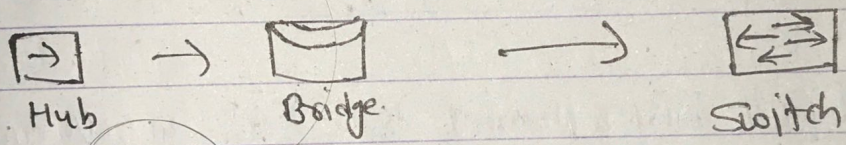
PC1 wants to send some traffic to PC2. To do so, it needs to know PC2's MAC. So, PC1 send ARP req frame (broadcast). Now SW1 will flood it from all int. except from where it received. SW2 & SW3 also do same. PC2 replies with unicast ARP reply But problem comes here? The broadcast frames circulates from SW1, SW2 SW3 continuously. Also, in layer 2 there is no TTL. So, they go in infinite loop. If enough of these looped broadcasts accumulate in network, the network will be too congested for legitimate traffic to use the network. This is called a **broadcast storm**.

Network congestion isn't the only problem. Each time a frame arrives on a switchport, the switch uses the source MAC address to learn the MAC and update its MAC add. table. When frame with same SC MAC add. repeatedly arrive on different interfaces, the switch is continuously updating the interface in the MAC add. table. This is known as **MAC Address Flapping**

The cost of int. on root bridge on is D.

Spanning tree protocol

- 'Classic' Spanning Tree Protocol' is IEEE 802.1D
- Switches from ALL vendors run STP by default.
- STP prevents layer 2 loops by placing redundant ports in a blocking state, essentially disabling the interface.
- These interfaces acts as backups that can enter a forwarding state if an active (= currently forwarding) interface fails.
- Interfaces in a forwarding state behave normally. They send and receive all normal traffic.
- Interface in a blocking state only send or receive STP mssg (called BPDU = Bridge Protocol data unit)



Spanning Tree Protocol still uses the term 'bridge'. However when we use the term 'bridge' we mean 'switch'. Bridges aren't used in modern network.

- By selecting which ports are a forwarding & which ports are blocking STP creates a single path to/from each point in network. This prevents layer 2 loops.
- There is a set process that STP uses to determine which ports should be in forwarding and which should be in blocking
- STP-enabled switches send/receive 'Hello BPDUs' out of all interfaces, the default timer is 2 sec (the switch will send Hello BPDU out of every interface, every 2 seconds).

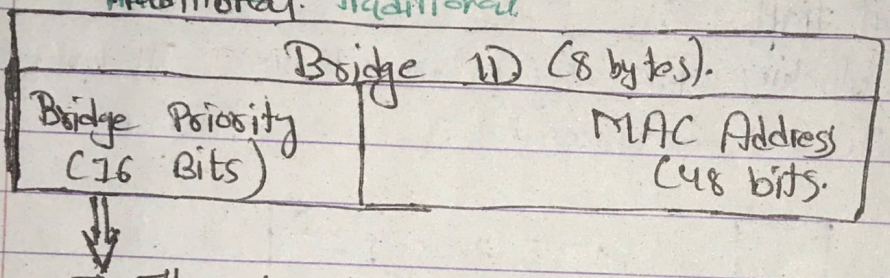
Scanned with CamScanner

If a switch receives a Hello BPDU on interface, it knows that int is connected to another switch (routers, PCs etc don't use STP, so they don't send Hello BPDUs)

Root Bridge, Bridge ID

Switches use one field in STP BPDU, the Bridge ID field to elect root bridge for the network.

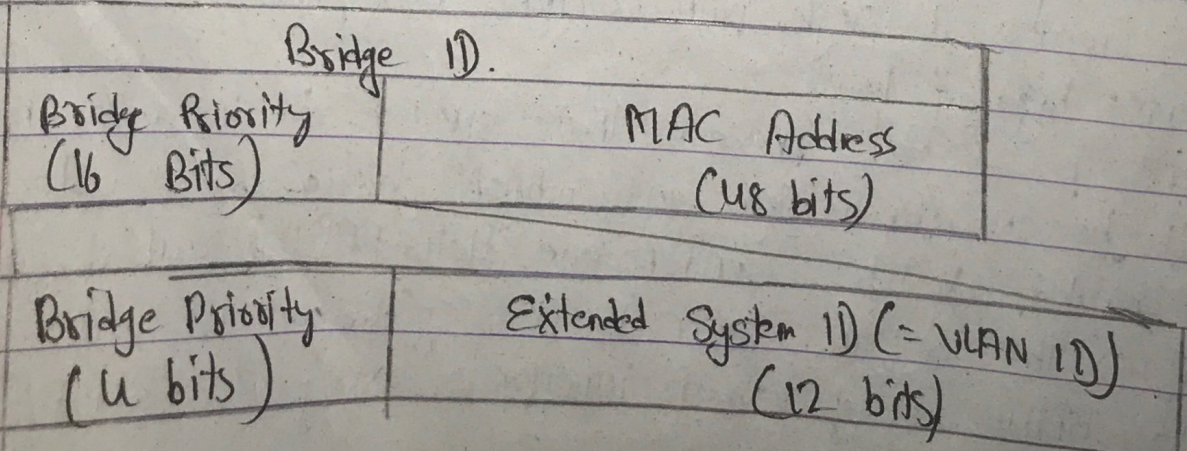
Traditional. Traditional



The default bridge priority is 32768 on all switches, so by default the MAC add is used as the tie-breaker (lowest MAC add. becomes root bridge)

The bridge ID is compared, if they tie MAC add is used as tie-breaker.

Updated Bridge ID



CISCO switches use a version of STP called PVST (Per-VLAN STP)
PVST runs a separate STP ('instance') in each VLAN, so in each VLAN different interfaces can be forwarding/blocking.
On the default VLAN of 1, the default bridge priority is actually 32769 (32768 + 1).

Root bridge, ~~select~~ BPDUs

- All interfaces on the root bridge are designated ports. Designated ports are in forwarding state.
- When a switch is on, it assumes it's the root bridge.
- It will only give up its position if it receives a BPDU from switch with lower bridge ID.
- Once the topology has converged and all the switched agree on the root bridge, only the root bridge sends BPDUs.
- Other switches in the network will forward these BPDUs, but won't generate their own original BPDUs

Root cost, root port

- ~~The switch with lowest~~
- The root cost is the total cost of the outgoing interfaces along the path to the root bridge.
Each interface has an associated spanning tree 'cost'.

Speed	STP Cost
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2

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To change cost, go to int-config mode then,

Spanning-tree vlan vlan_number cost your_preferred_cost

Spanning-tree vlan vlan_number port_priority your_preferred_priority

The ports connected to another switch's root port must be designated. Because the root port is the switch's path to the root bridge. another switch mustn't block it.

STP Port ID = port priority (default 128) + port no.

Spanning-tree vlan 1 cost 100

STP Process.

1) One switch is elected as the root bridge. All ports on the root bridge are designated ports (forwarding state). Root bridge selection.

a). Lowest bridge ID

(see priority, if same priority, lower MAC wins)

2). Each remaining switch will select ONE of its interface to be its root port (forwarding state). Ports across from the root ports are always designated ports.

Root port selection

a) Lowest root cost

b) Lowest neighbour bridge ID

c). Lowest neighbour port ID.

3) Each remaining collision domain will select ONE interface to be a designated port (forwarding state). The other port in collision domain will be non-designated (blocking)

Designated port selection

a) Interface on switch with lowest root cost.

b) Interface on switch with lowest bridge ID

Some STP Commands.

All commands run from privileged exec mode.

① Show spanning-tree or show span tree

If we want to see STP of particular VLAN.

② show span vlan 1.

If we wanna see it detailly use the command.

③ show span tree detail
④ show spanning-tree summary

STP Protocol Part(2). 21 Nov.

Spanning Tree Port States.

STP Port State	Stable / Transitional
Blocking	Stable
Listening	Transitional
Learning	Transitional
Forwarding	Stable

(Disabled)

- Root / Designated Ports remain stable in a Forwarding State.
- Non-designated ports remain stable in blocking state.
- Listening & learning are transitional states which are passed through when interface is activated or when a blocking port must transition to Forwarding state due to change in network topology.

- There is one more state disabled state: It means administratively shut down the interface.

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a) Blocking state.

- Non-designated ports are in blocking state.
- Interfaces in blocking state are effectively disabled to prevent loops.
- Interfaces " " don't send/receive regular network traffic.
- " " " receive STP BPDUs.
- " " " " don't forward STP BPDUs.
- " " " " donot learn MAC address.

b) Listening state.

- After the blocking state, interfaces with the Designated or Root role entering the Listening state.
- Only designated or root ports enter the listening state (Non-designated ports are always blocking).
- The listening state is 15 seconds long by default. This is determined forward delay timer.

- An interface in listening state ONLY, forwards/receives STP BPDUs.
- " " " " " doesn't send/receive regular traffic.
- " " " " doesn't learn MAC addresses from regular traffic that arrives on the interface.

c) Learning state.

- After the ~~learning~~ listening state, a designated or root port will enter the Learning state.

- The Learning state is 15 seconds long by default. This is also determined by forward delay timer.

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- An interface in the learning state ONLY sends/receives STP BPDUs
- An interface in the learning state doesn't send/receive regular traffic.
- An interface in the learning state learns MAC addresses from regular that arrives on interface.

d) Forwarding state.

- Root and Designated ports are in Forwarding state.
- A port in the Forwarding state operates as normal.
- A port in the Forwarding state sends/receives BPDUs.
- A port in the " " sends/receives normal traffic.
- A " " " learn MAC address.

Spanning Tree Timers

STP Timer	Purpose	Duration
Hello	How often the root bridge sends hello BPDUs	2 sec.
~~Forward delay~~ ~~Max Age~~		
Forward delay	How long the switch will stay in the listening & learning states (each state is 15 sec = total 30 sec)	15 sec.
Max Age	How long an interface will wait after ceasing to receive Hello ~~BPDUs~~ BPDUs to change the STP topology.	20 sec (10* hello)

The switches willy only forward BPDUs on their DESIGNATED PORTS.

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c) Max Age

- If another BPDU is received before the max age timer counts down to 0, the time will reset to 20 sec, and no changes will occur.

- If another BPDU is not received, the max age timer counts down to 0 and the switch will re-evaluate its STP choices, including root bridges, ~~and local bridges.~~ and local root, designated & non-designated ports.

- If a non-designated port is selected to become a designated or root port, it will transition from blocking state to listening state (15 seconds), learning state (15 seconds) and then finally the forwarding state. So, it can take a total of 50 seconds for a blocking interface to transition to forwarding.

- These timers and transitional states are to make sure that loops aren't accidentally created by an interface moving to forwarding state too soon.

NOB: A forwarding & interface can move directly to a blocking state (there is no worry about creating a loop by blocking an interface).

A blocking interface cannot move directly to forwarding state, it must go through learning & listening states.

NOP Things to remember for exam.

- Cisco PVST+ uses the destination MAC add. of 0100.0ccc.cccd for its BPDUs.

PVST = Only ISL trunk encapsulation
PVST+ = Supports 802.1Q.

- Regular STP uses a destination MAC address of 0180.c200.0000.

- The STP timers on root bridge determine the STP timers of the entire network.

Spanning Tree Optional Features (STP Toolkit)

1) PortFast:

Interfaces connected to end hosts are designated. However when they are 1st connected to PCs, they must go through listening & learning states first before they can start forwarding traffic. So, it take 30 sec before they can start forwarding. Even connected to endhosts doesn't have risks of forming layer 2 loops. So, its just loss of 30 secs.

Portfast allows a port to move immediately to Forwarding state, bypassing Listening & Learning.

If used, it must be enabled only on ports connected to end hosts.

If enabled on ports connected to another switch it could cause a layer 2 loop.

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PortFast commands:

Sw1 # int g0/2
Sw1(config-if) # spanning-tree ~~portfast~~ portfast

We can also enable portfast by following commands:
Sw1(config)# spanning-tree portfast default.
This enables portfast on all access ports (not trunk ports)

But there can be risk, if some employee connected portfast enabled port switch to another switch. For this we have use another feature BPDU Guard.

2) BPDU Guard

If an int with BPDU guard enabled receives a BPDU from another switch, the interface will shut down to prevent a loop from forming.

To enable BPDU guard on interface.

Sw 1 (config) # int g0/2
Sw 1 (config-if) # spanning-tree bpduguard enable
Sw 1 (config-if) #

There is also an option to enable it by default.

Sw1(config)# spanning-tree portfast bpduguard default

This enables BPDU ~~guard~~ Guard on all portfast-enabled interfaces.

3)# Root Guard

If we enable root guard on an int., even if it receives a superior BPDU (lower bridge ID) on that interface, the switches won't accept the new switch as root bridge. The interface will be disabled.

Loop Guard.

If you enable loop guard on an interface, even if the interface stops receiving BPDUs, it won't start forwarding. The interface will be disabled.

NOTE: We probably don't have to know these STP features for CCNA. But we need to know Portfast and BPDU Guard.

Configuring Spanning Tree Mode.

spanning-tree mode ?

* mst
* pvst
* rapid-pvst

Modern Cisco switches run rapid-PVST by default, and usually there is no reason to change it.

Configuring the Primary root bridge

We can configure root bridge as well as secondary root bridge, which is next-in line to become root bridge if the current root bridge fails.

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Sw2# spanning-tree vlan 1 root primary

The spanning-tree vlan vlan-number root primary command sets the STP priority to 24576. If another switch already has a priority lower than 24576, it sets this switch's priority to 4096 less than other switch's priority.

~~Sw2#~~

Sw2# spanning-tree vlan 1 root secondary

The 'spanning-tree vlan vlan-number root secondary' command sets the STP priority to 28672.

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